Protein Structure Modelling

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1. Results

1.1 Multiple Sequence Alignment (MSA)

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CLUSTAL O(1.2.4) multiple sequence alignment


Ma07            MVDLTNTIFGIVSILIVICIIIAIVLTANNKIHELTFEQKAIPKFSIDLQFDPYPQVVNI      60
Baybay          MVDLTNTIFGIVSILIVICIIIAIVLTANNKIHELTFEQKAIPKFSIDLQFDPYPQVVNI      60
Korean          MIDLTNTIFGIVSILIVICIIIAIVLTSNNKIHELTFEQKAIPKFSIDLQFDPYPQVVNI      60
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Ma07            ENQYDCNVHSLRVCDINDSTTLFGCKELIVRCHHFDEDTPFIENNETLTIPKNQTADEGY      120
Baybay          ENQYDCNVHSLRVCDINDSTTLFGCKELIVRCHHFDEDTPFIENNETLTIPKNQTADEGY      120
Korean          ENQYDCNVHSLRVCDINDSTTLFGCKELIVRCHHFDEDTPFIENNETLTIPKNQTADEGY      120
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Ma07            ALAITTISDACNPYHGDMTLVTANSDASEYMLICTCKNPGYIGNESLLGNCTSVFICNGK      180
Baybay          ALAITTISDACNPYHGDMTLVTANSDASEYMLICTCKNPGYIGNESLLGNCTSVFICNGK      180
Korean          ALAITTISDACNPYHGDMTLVTANSDASEYMLICTCKNPGYIGNESLLGNCTSVFICNGK      180
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Ma07            VDDVNKPLNEINCVCESRQKTIRYDDGLPVCKDLLVHEANELYDDWSNIVPWSSFRQLET      240
Baybay          VDDVNKPLNEINCVCESRQKTIRYDDGLPVCKDLLVHEANELYDDWSNIVPWSSFRQLET      240
Korean          VDDVNKPLNEINCVCESRQKTIRYDDGLPVCKDLLVHEANELYDDWSNIVPWSSFRQLET      240
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Ma07            SNFNPTISGNLKTSRLLDPCRNSIHDTSIEILNASYDSVRKQCSLTDNGFPVVNDLLRFS      300
Baybay          SNFNPTISGNLKTSRLLDPCRNSIHDTSIEILNASYDSVRKQCSLTDNGFPVVNDLLRFS      300
Korean          SNFNPTISGNLKTSRLLDPCRSSIHDTSIEILNASYDSVRKQCSLTDNGFPVVNGLLRFS      300
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Ma07            STEKAYVGASAVLATGEYEFIRFSDNLAGQRHIIGIVSRGLKFSDKYKNTTTVVHPTNGL      360
Baybay          STEKAYVGASAVLATGEYEFIRFSDNLAGQRHIIGIVSRGLKFSDKYKNTTTVVHPTNGL      360
Korean          STEKAYVGASAVLATGGYEFIRFSDNLAGQRHIIGIVSRGLKFSDKYKNTTTVVHPTNGL      360
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Ma07            SLGHSTALSVKSKTKFVGPACYGSWPTYSCSSIEERDSYITSGLALPRGRVCPGAFLWSR      420
Baybay          SLGHSTALSVKSKTKFVGPACYGSWPTYSCSSIEERDSYITSGLALPRGRVCPGAFLWSR      420
Korean          SLGHSTALSVKSKTKFVGPACYGSWPTYSCSSIEERDSYITSGLALPRGRVCPGAFLWSR      420
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Ma07            SDWDDHEFLVVRSATQNIQDGYSLSTTKLTSLDDIDSYGIQWSTQTSSEYTGLLRFLNSA      480
Baybay          SDWDDHEFLVVRSATQNIQDGYSLSTTKLTSLDDIDSYGIQWSTQTSSEYTGLLRFLNSA      480
Korean          SDWDDHEFLVVRSATQNIQDGYSLSTTKLTSLDDIDPYGIQWSTQTSSEYTGLLRFLNSA      480
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Ma07            DYRIHKQTITG 491
Baybay          DYRIHKQTITG 491
Korean          DYSIHKQTITG 491
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Figure 1: Sequence Alignment

The alignment reveals that the Ma07 and Baybay Strains are highly conserved, sharing identical amino acid sequences across nearly the entire 491-residue length, whereas the Korean strain exhibits several distinct non-synonymous changes that suggest it is an evolutionary outlier.

Table 1. Mutation Table Summary

Table 1: Mutation Table Summary

Position	Ma07	Korea	Baybay	Remarks
2	V	I	V	Korean-specific change
28	A	S	A	Korean-specific change

This pattern suggests that while the fundamental biological function of the protein is preserved across all three strains, the Korean strain has undergone specific selective pressures or genetic drift that differentiated its surface-exposed or flexible domains from the Mao07 and Baybay variants.

1.2 Final 3D Protein Structure Models (Highest-Confidence Only)

1.2.1 Ma07 Isolates

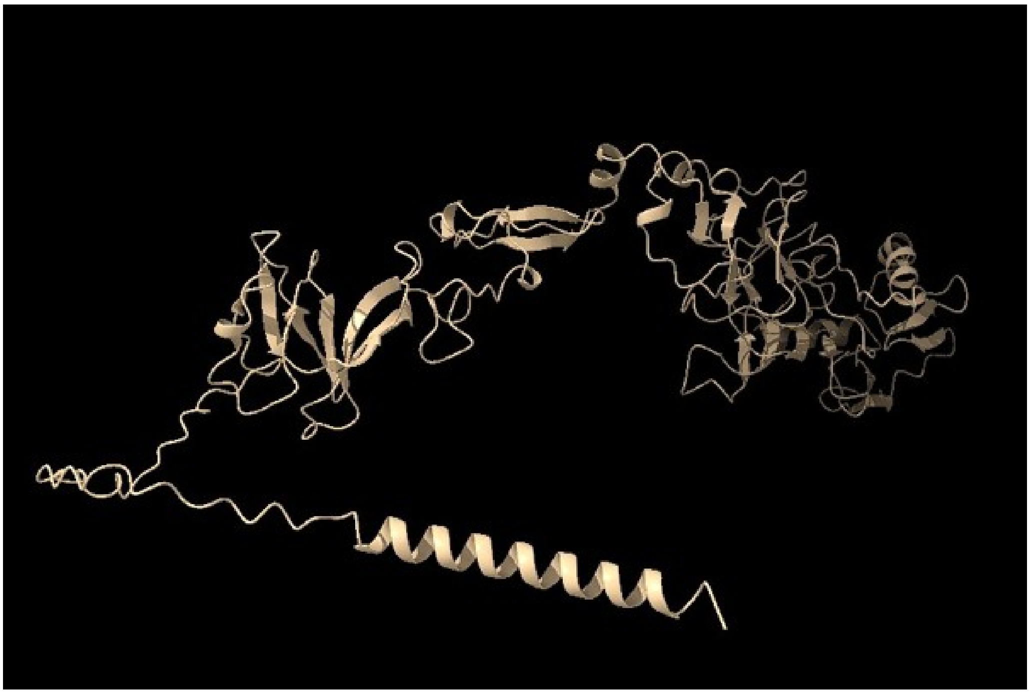

Figure 2: Ma07 Isolate Model rank 001

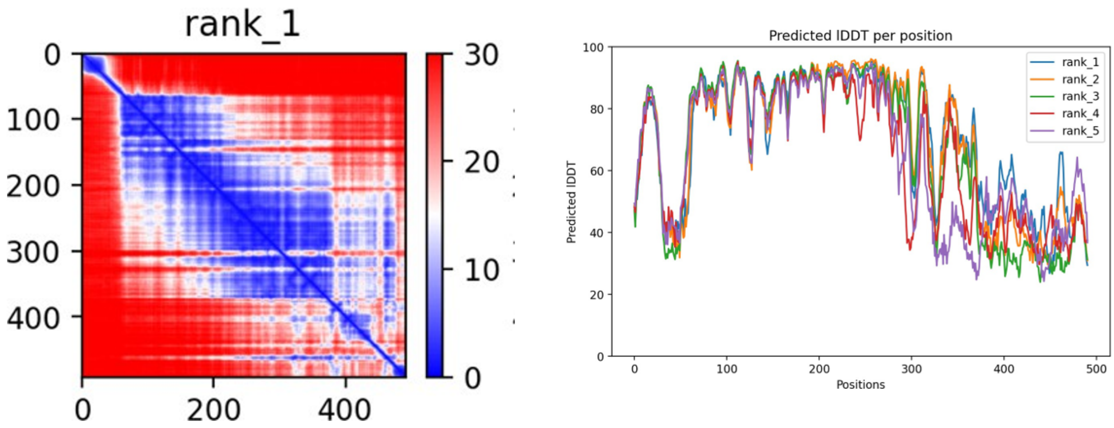

Figure 3: *Ma07 Isolate Model PAE and PLDDT models*

1.2.2 Korean Isolates

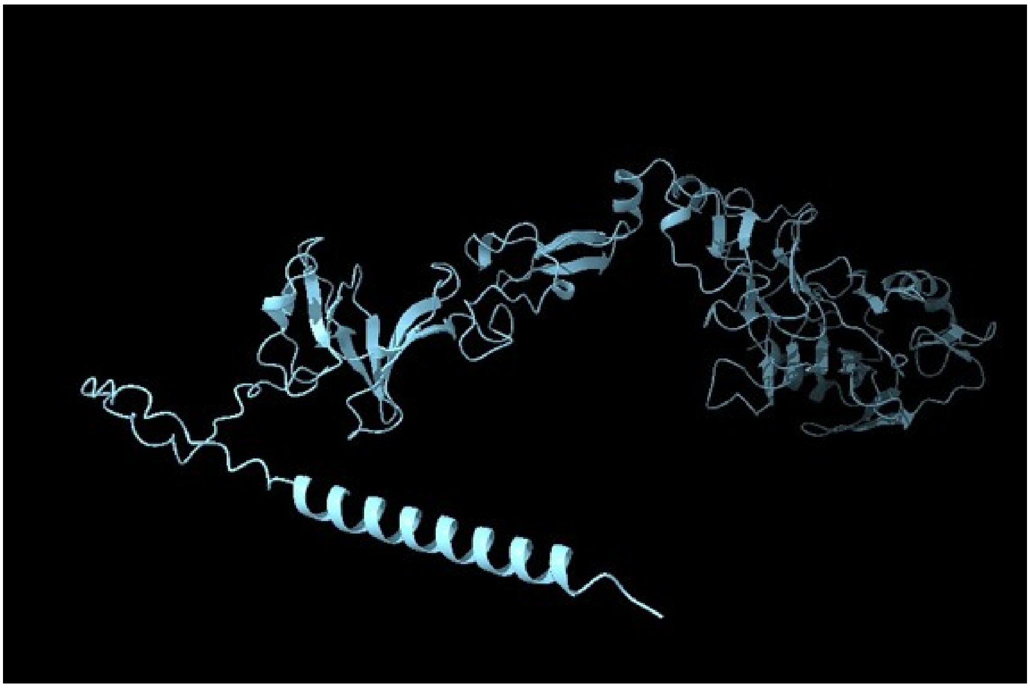

Figure 4: Korean Isolate Model rank 001

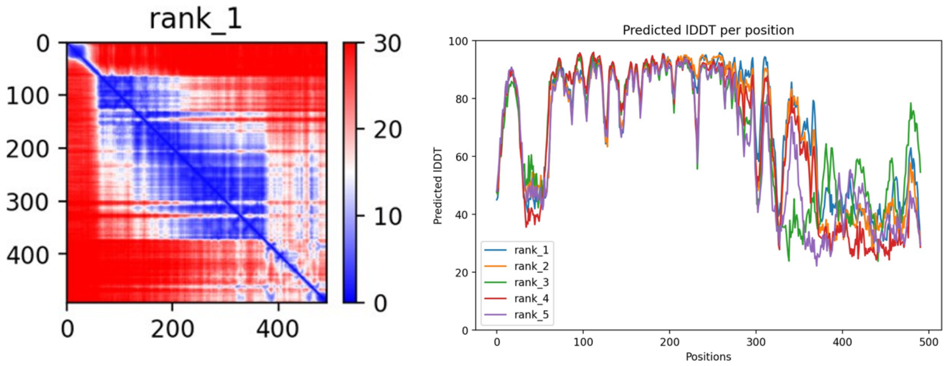

Figure 5: Korean Isolate Model PAE and PLDDT models

1.2.3 Baybay Isolates

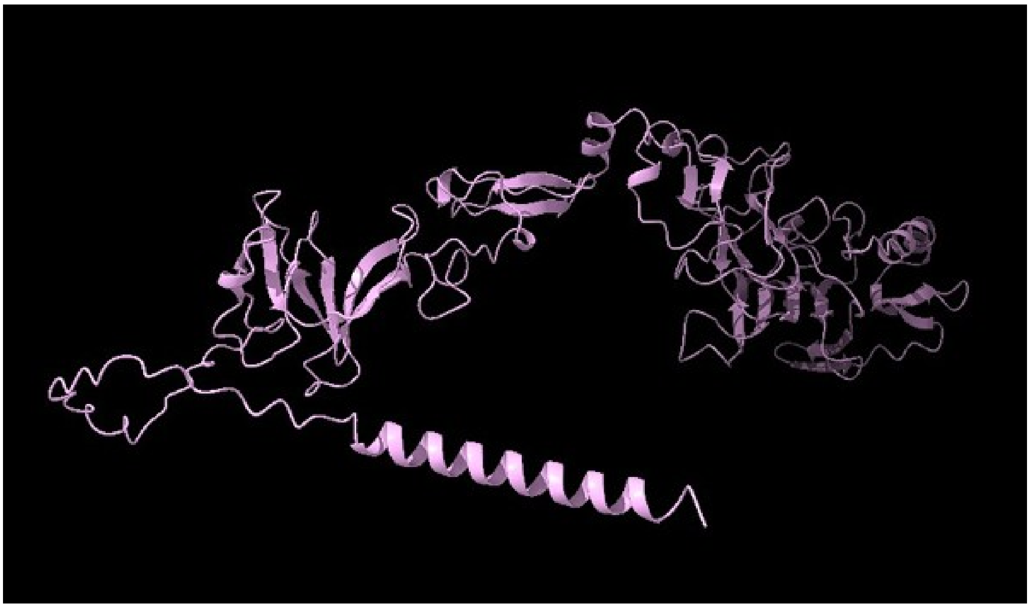

Figure 6: Baybay Isolate Model rank 001

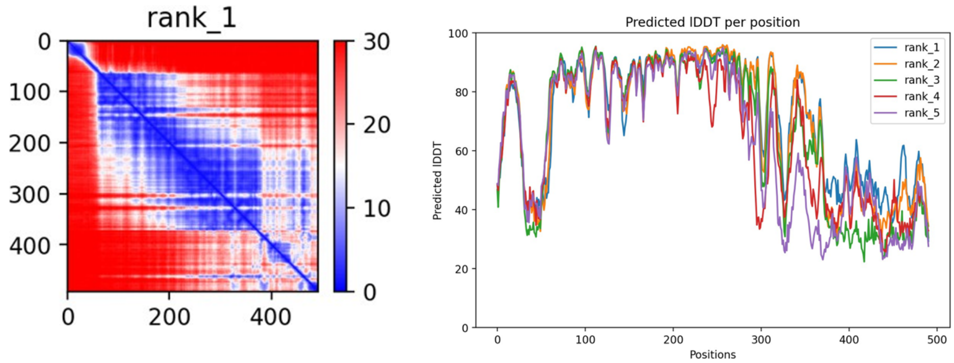

Figure 7: Baybay Isolate Model PAE and PLDDT models

2. Structural Analysis Figures

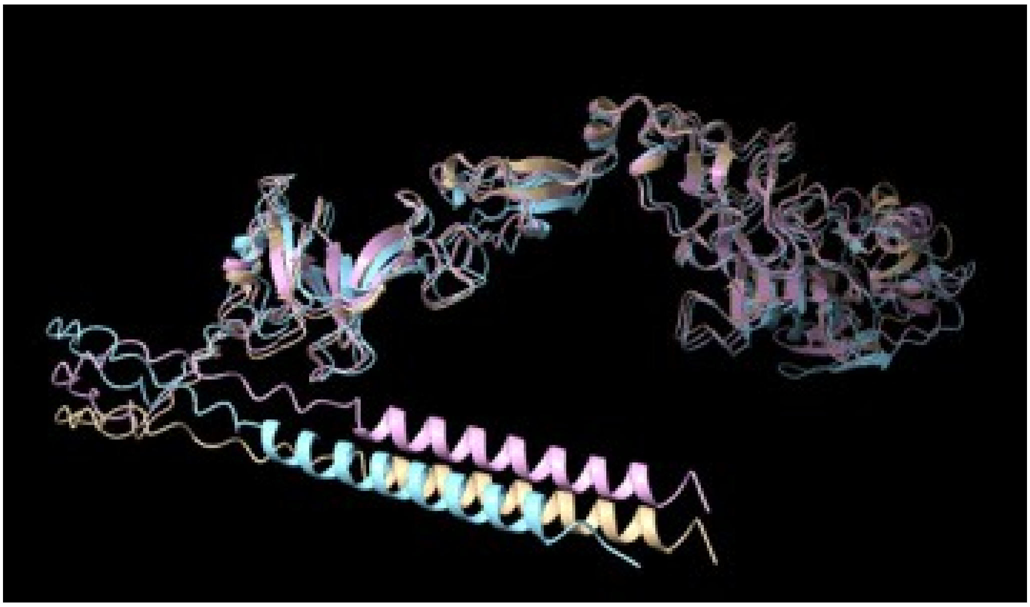

Figure 8: Overlaid ribbon structures of the three isolates

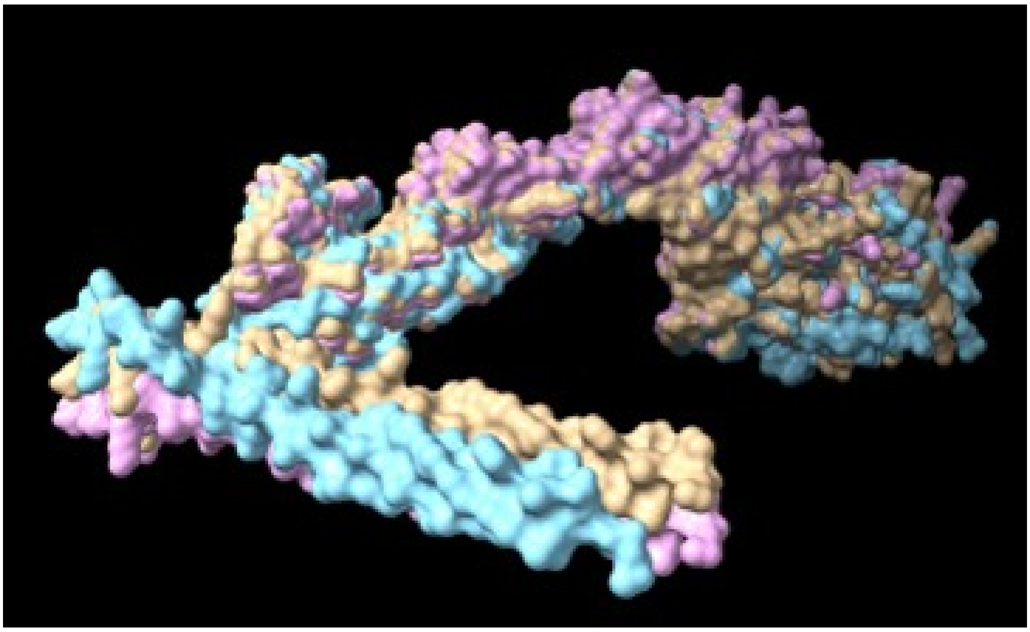

Figure 9: Overlaid surface structures of the three isolates

The structural overlay found in Figures 8 and 9 shows the Ma07 (yellow), Korean (blue), and Baybay (pink) isolates, providing a visual confirmation of their extreme structural conservation, especially within the central core of the protein. The overlay highlights that while geographic divergence exists at the sequence level, maintaining a functional 3D shape has kept the structural integrity of these isolates consistent.

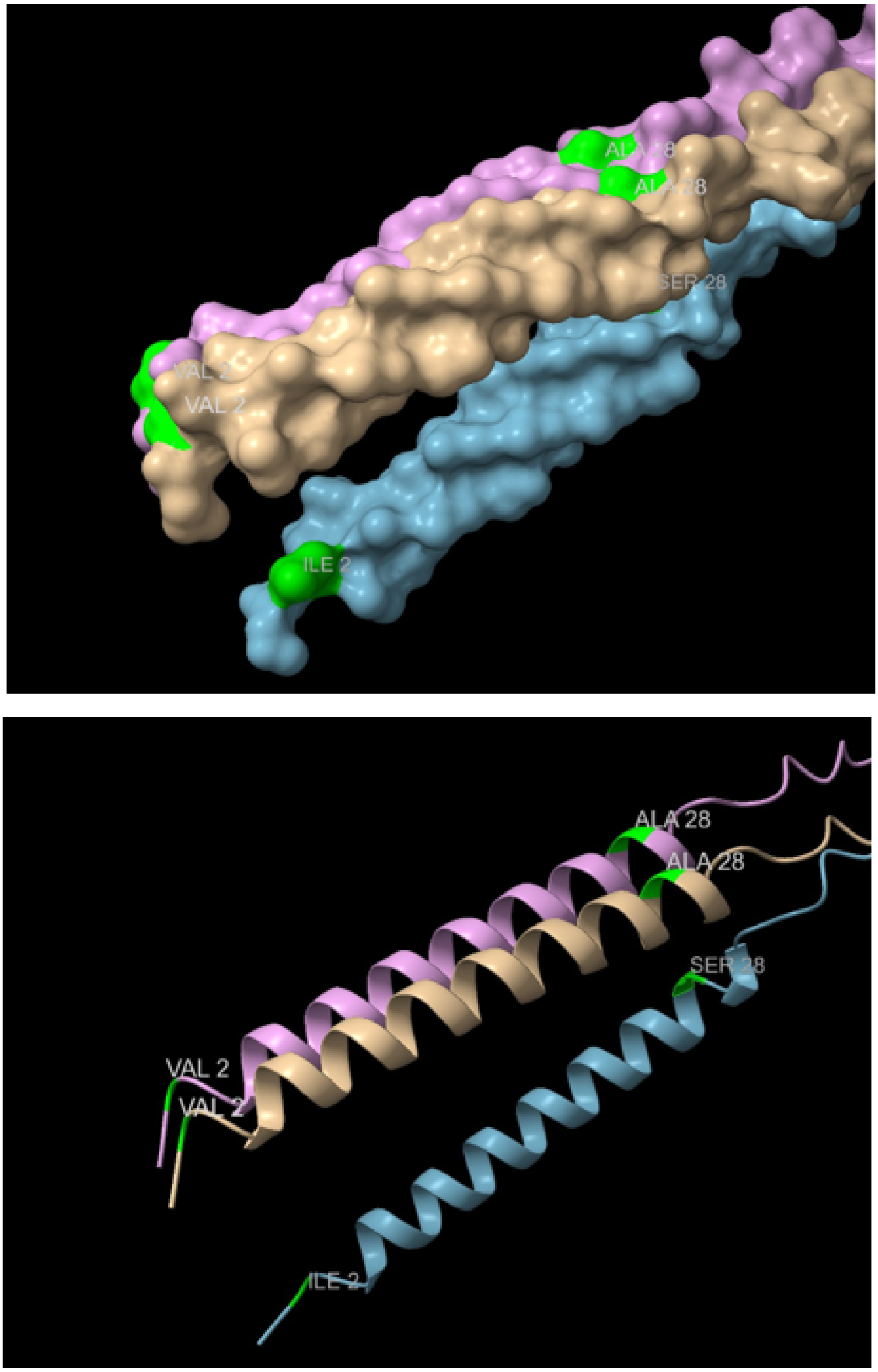

Figure 10: Mutations are highlighted both in ribbon and surface structures

The 3D structural analysis of the three isolates is presented through an overlay of their ribbon and surface models. Within these structures, two specific mutation sites are highlighted in lime green to demonstrate regional divergence, where at position 2, the valine residue found in Ma07 and Baybay is substituted with isoleucine in the Korean isolate, and at position 29, the Korean isolate features a serine residue in place of alanine. These localized changes are visualized in both detailed ribbon views and expanded surface structures to assess their impact on the protein's overall topography.

2.1 Electrostatic Surface Analysis

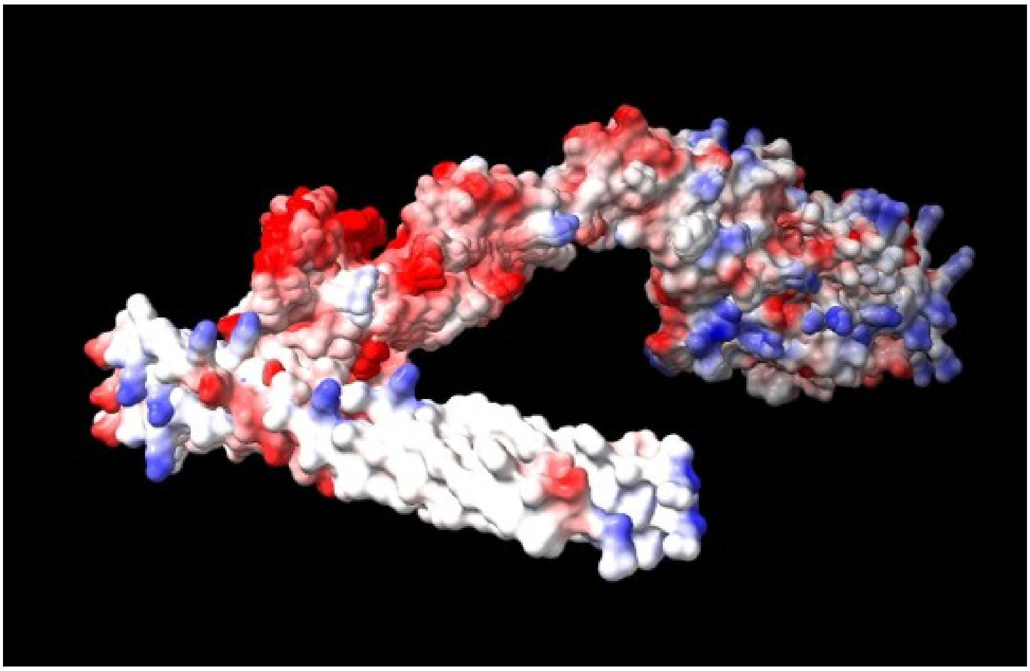

Figure 11: Overlaid electrostatic comparison of the three isolates

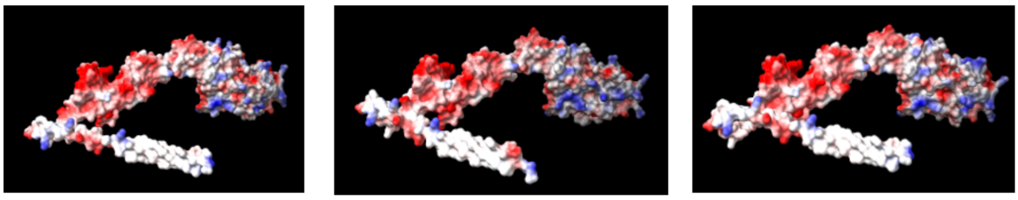

Figure 12: Electrostatics of the three isolates: Ma07 (left), Korean (middle), and Baybay (right)

3. Discussion

Mutation Table Summary

Table 2: Mutation Table Summary

Position	Ma07	Korea	Baybay	Remarks
2	V	I	V	Korean-specific change
28	A	S	A	Korean-specific change

The multiple sequence alignment and structural overlay of the pif-1 envelope protein reveal that the Ma07 and Baybay isolates are nearly identical, suggesting a shared evolutionary origin or low selective pressure within their respective regions. In contrast to the Korean isolate, it represents a distinct genetic variant within the *Oryctes rhinoceros* nudivirus (OrNV) population.

OrNV is a critical biocontrol agent used to manage the coconut rhinoceros beetle (CRB) across tropical Asia and the Pacific (Russo, 2024). Within the Nudiviridae family, pif-1 (Per Os Infectivity Factor 1) is a specialized envelope protein that forms part of the core complex required for the virus to initiate infection in the beetle's midgut epithelium (Peterson, 2025).

The conservation observed across all isolates suggests that those regions are vital for the structural stability of the viral envelope, where any significant mutation could compromise the virus's ability to persist in the agricultural environment.

The structural analysis using AlphaFold2 highlights how localized mutations in the Korean isolates may influence the virus's efficacy as a biocontrol tool. Specifically, the Alanine to Serine mutation introduces a polar hydroxyl group into the N-terminal region, potentially altering the protein's interaction with the alkaline environment of the beetle's midgut.

Furthermore, the Serine to Proline mutation acts as a "helix-breaker," likely increasing the flexibility of the C-terminal tail. Because PIF proteins must work in concert to facilitate the fusion of the viral envelope with the host cell membrane, such structural changes in the Korean strain could lead to variations in "host-kill" speed or infectivity rates compared to the Ma07 and Baybay variants commonly found in different palm-growing regions (Bézier et al., 2015).

The electrostatic surface analysis further distinguishes these isolates, showing a reduction in surface charge for the Korean strain. In agricultural entomology, the binding affinity of nudiviruses to the larval midgut is often governed by electrostatic interactions between viral proteins and the peritrophic matrix (Peterson et al., 2022).

These shifts in the Korean PIF-1 suggest a potential adaptation to specific host biotypes or environmental conditions unique to its geographical range.

AlphaFold2 provided the necessary high-resolution confidence (pLDDT) to interpret these terminal regions accurately, as traditional homology models often lack the specific experimental templates required for the diverse and rapidly evolving proteins found in insect viruses like OrNV (Jumper et al., 2021).

References

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5. Petersen, J. M., Bézier, A., Drezen, J. M., & van Oers, M. M. (2022). The naked truth: An updated review on nudiviruses and their relationship to bracoviruses and baculoviruses. *Journal of Invertebrate Pathology*, 189, 107718.

6. Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., et al. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873), 583–589.